

Foundation (Jalapeno)

- Q1.** Given $f(x) = x + 2$ and $g(x) = 2x - 1$, what is $f(g(x))$?
(A) $2x + 1$
(B) $2x + 3$
(C) $2x - 1$
(D) $2x + 2$
(E) $2x - 3$
- Q2.** If $f(x) = \frac{1}{x}$ and $g(x) = x - 1$, what is the domain of $f(g(x))$?
(A) $(-\infty, 1) \cup (1, \infty)$
(B) $(-\infty, 1)$
(C) $(1, \infty)$
(D) $[1, \infty)$
(E) $(-\infty, 1] \cup [1, \infty)$
- Q3.** Given $f(x) = x^2$ and $g(x) = \sqrt{x}$, what is $f(g(x))$?
(A) x
(B) x^2
(C) $\sqrt{x}$
(D) x^4
(E) $\sqrt{x^2}$
- Q4.** If $f(x) = 2x$ and $g(x) = x - 3$, what is $g(f(x))$?
(A) $2x - 3$
(B) $2x + 3$
(C) $x - 3$
(D) $x + 3$
(E) $2x - 6$
- Q5.** Given $f(x) = x^2 - 1$ and $g(x) = x + 2$, what is $f(g(x))$?
(A) $x^2 + 3x + 3$
(B) $x^2 + 4x + 3$
(C) $x^2 - 4x + 3$
(D) $x^2 + 4x - 1$
(E) $x^2 - 4x - 1$
- Q6.** If $f(x) = \frac{1}{x+1}$ and $g(x) = x - 2$, what is the domain of $f(g(x))$?
(A) $(-\infty, 3) \cup (3, \infty)$
(B) $(-\infty, 1)$
(C) $(1, \infty)$
(D) $(-\infty, 3]$
(E) $[3, \infty)$
- Q7.** Given $f(x) = |x|$ and $g(x) = x - 4$, what is $f(g(x))$?
(A) $|x - 4|$
(B) $x - 4$
(C) $x + 4$
(D) $|x| - 4$
(E) $|x + 4|$
- Q8.** If $f(x) = x^3$ and $g(x) = 2x$, what is $f(g(x))$?
(A) $8x^3$
(B) $8x$
(C) $2x^3$
(D) $2x$
(E) $2x^2$

Open Ended Questions (FRQ)

- Q9.** A city's population P (in thousands) can be modeled by $P(f(t))$, where $f(t) = t^2 + 1$ and t is the number of decades since 2000. If $P(x) = 10 + 5x$, find the population of the city 3 decades after 2000.

- Q10.** Given $f(x) = 2x - 1$ and $g(x) = x^2 + 1$, find the domain of $f(g(x))$. Show all steps and explain your reasoning.

Chef's Tips

- Remind students to check the domain of composite functions.
- Emphasize the importance of correct order of operations when evaluating composite functions.
- Encourage students to use real-world examples to illustrate the concept of composite functions.